Recape

Def

(*) A matrix $A_{n\times n}$ is said to be <u>invertible</u> if there exists a matrix B s.t. $AB = BA = I_n$. Then B is the inverse of A.

~~Prop (1)~~ $B = A^{-1} = \frac{1}{|A|}(\text{Adj } A)$

Prop: (1) Invers (of an invertible matrix) is ~~un~~ unique.

(2) A is invertible (or inverse of A exists) $\Longleftrightarrow |A| \neq 0$
i.e. A is non-singular

(3) $A_{n\times n}$, $B_{n\times n}$ are invertible $\Rightarrow (AB)^{-1} = B^{-1}(A)^{-1}$

(4) $(A^{-1})^{-1} = A$, where A is invertible.

(5) $(A^t)^{-1} = (A^{-1})^t$, "A" " .

(6) $(A_{n\times n})(B_{n\times n}) = 0 \Rightarrow \begin{cases} A = 0 & \text{when } |B| \neq 0 \\ B = 0 & \text{when } |A| \neq 0 \end{cases}$

zero matrix

Def: A (real) square matrix A is said to be symmetric if $A^t = A$ and skew-symmetric if $A^t = -A$.

Prop: (1) A, B are two symmetric matrices of the same order then

(a) $A+B$ is symmetric and

(2) (b) AB is symmetric $\Longleftrightarrow AB = BA$

(2) AA^t and A^tA are symmetric where $A_{m\times n}$ is a real matrix

(3) A is a square matrix $\Rightarrow A+A^t$ is symmetric and $A-A^t$ is skew-symmetric.

(4) A real (or complex) square matrix can be uniquely expressed as the sum of a symmetric matrix and a skew-symmetric matrix

---

Inverse of a square matrix

(1) Let A be square matrix, $|A| \neq 0$ and $AB = I$. Then $B = A^{-1}$

Pf: $|A| \neq 0 \Rightarrow A^{-1}$ exists.

Let C be left inverse of A then $CA = I \rightarrow (1)$

Now $AB = I \Rightarrow CAB = CI \Rightarrow IB = CI \Rightarrow B = C$

$\therefore AB = BA = I \Rightarrow B = A^{-1}$.

(2) Let A be a nonsingular square matrix and $BA = I$.

Then $B = A^{-1}$.

Orthogonal Matrices

Def A real $n \times n$ matrix is said to be <u>orthogonal</u> if $AA^t = I_n$.

Prop 1: A is orthogonal $\Rightarrow |A| = \pm 1$

Sol^n:- $AA^t = I \Rightarrow |AA^t| = |I| = 1$

$$\Rightarrow |A||A^t| = 1$$

$$\Rightarrow |A|^2 = 1 \Rightarrow |A| = \pm 1$$

Prop 2: A is orthogonal $\Rightarrow A^{-1} = A^t$

Sol^n:- $AA^t = I \longrightarrow (1)$ $\because A$ is square matrix and $|A| \neq 0$. Then $(1) \Rightarrow A^{-1} = A^t$.

$$\Rightarrow AA^tA = IA \Rightarrow A(A^tA) = AI$$

$$\Rightarrow A(A^tA - I) = 0$$

$$\Rightarrow A^tA - I = 0 \quad (\because |A| \neq 0)$$

$$\Rightarrow A^tA = I \longrightarrow (2)$$

(1) & $(2) \Rightarrow A^{-1} = A^t$

Prop 3: $A_{n\times n}$, $B_{n\times n}$ are orthogonal $\Rightarrow AB$ is orthogonal

Sol^n:- $(AB)(AB)^t = (AB)(B^tA^t) = A(BB^t)A^t = (AI_n)A^t$

$$= AA^t = I_n$$

Prop 4: A is orthogonal $\Rightarrow A^{-1}$ is orthogonal

Sol^n:- $A^{-1}(A^{-1})^t = A^{-1}(A^t)^{-1} = A^{-1}(A^t)^{-1}$

$$= (A^tA)^{-1} = I^{-1} = I$$

Complex Matrices

Def: A matrix A whose elements are taken from the field $\mathbb{C}$ is said to be a complex matrix.

A can be expressed as $A = P + iQ$, where P, Q are real matrices.

$\bar{A} = P - iQ$ is said to be the conjugate of A.

~~The elements of $\bar{A}$ are the~~

(i, j) th element of A is a_{ij}

$\Rightarrow (i, j)$-th " " $\bar{A}$ is $\bar{a}_{ij}$

Conjugate Transpose (transposed conjugate) of A is $(\bar{A})^t \equiv A^*$

Prop 1: $\bar{\bar{A}} = A$ (obvious)

Prop 2: $\overline{AB} = \bar{A}\,\bar{B}$ (A, B are complex matrices & AB is defined)

Sol^n:- Let $A = P + iQ$ and $B = R + iS$, where P, Q, R, S are real matrices.

HW Compute AB, $\overline{AB}$, $\bar{A}$, $\bar{B}$, $\bar{A}\bar{B}$ and show that $\overline{AB} = \bar{A}\bar{B}$

Prop 3:- $(\bar{A})^t = \overline{(A^t)}$

Sol^n: ~~Hints~~ HW $A = P + iQ$, where P, Q are real matrices

$\Rightarrow \bar{A} = P - iQ$, $A^t = P^t + iQ^t$

$\Rightarrow (\bar{A})^t = P^t - iQ^t = \overline{(A^t)}$

Prop 4: A, B are complex matrices.

~~Soln~~ :- ~~HW~~ (i) $(A^*)^* = A$

(ii) $(cA)^* = \bar{c} A^*, \quad c \in \mathbb{C}$

(iii) $(A+B)^* = A^* + B^*$ ($A+B$ is defined)

(iv) $(AB)^* = B^* A^*$ (when AB is defined)

Soln. – HW.

Prop 5: – If A be a non singular complex matrix, then $(A^*)^{-1} = (A^{-1})^*$

Soln:– Since A is non singular A^{-1} exists.

Then $AA^{-1} = A^{-1}A = I_n$ (n is the order of A)

$(I_n)^* = (AA^{-1})^* = (A^{-1}A)^*$

$\Rightarrow I_n = (A^{-1})^* A^* = A^*(A^{-1})^*$

~~$\Rightarrow (A^{-1})^*$ is the inverse~~

From the definition of an inverse and its uniqueness it follows that $(A^{-1})^*$ is the inverse of A^*.

$\therefore (A^*)^{-1} = (A^{-1})^*$.

Now $|A| \neq 0 \Rightarrow |A^*| \neq 0 \Rightarrow (A^*)^{-1}$ exists.

Now $(A^*)(A^{-1})^* = (A^{-1}A)^* = I$

$\Rightarrow (A^*)^{-1} = (A^{-1})^*$

Hermitian & Skew-Hermitian matrices

Def: A complex square matrix A is said to be Hermitian if

$$A^* = A$$

and skew-Hermitian if $A^* = -A$, where A^* ~~is the~~ $= (\bar{A})^t$

Ex- $\begin{bmatrix} 1 & 2+i & -1 \\ 2-i & 2 & i \\ -1 & -i & 0 \end{bmatrix}$, $\begin{pmatrix} 1 & 2 & 3i \\ 2 & 0 & 4 \\ -3i & 4 & 2 \end{pmatrix}$ are Hermitian

$\begin{pmatrix} 0 & 2 & i \\ -2 & 0 & 1+i \\ i & -1+i & 0 \end{pmatrix}$, $\begin{pmatrix} 0 & 2i & 1+2i \\ 2i & i & 1 \\ -1+2i & -1 & 0 \end{pmatrix}$ are skew-Hermitia

Observe the diagonals of these matrices.

Th: If $H = P + iQ$ be a Hermitian matrix then

(i) the diagonal elements of H are reals,

(ii) P is a real symmetric matrix and Q is a skew-~~sym~~ symmetric matrix.

Pf: (i) Let $H = (h_{ij})$. Then $\overline{h_{ij}} = h_{ij} \Rightarrow \overline{h_{ii}} = h_{ii}$
$\Rightarrow h_{ii}$ is a real.

(ii) H is Hermitian $\Rightarrow (\bar{H})^t = H$

$$\Rightarrow (P - iQ)^t = P + iQ$$

$$\Rightarrow P^t - iQ^t = P + iQ$$

$$\Rightarrow P^t = P \text{ and } Q^t = -Q \quad \square$$

<u>Th</u>: If $S = M + iN$ be a skew-symmetric matrix, then

(i) the diagonal elements of S are purely imginary ~~or~~ ~~and~~ numbers or zero,

(ii) M is a real skew-symmetric matrix and N is a real symmetric matrix.

<u>Pf</u>: (i) Let $S = (s_{ij})$. Then $\bar{s}_{ij} = -s_{ji} \Rightarrow \bar{s}_{ii} = -s_{ii}$

$\Rightarrow s_{ii}$ is purely imaginary or zero.

(ii) S is skew-Hemitian $\Rightarrow \overline{(M+iN)}^t = -(M+iN)$

$$\Rightarrow (M - iN)^t = -(M + iN)$$

$$\Rightarrow M^t - iN^t = -M - iN$$

$$\Rightarrow M^t = -M \text{ and } N^t = N \quad \square$$

<u>Th</u>: If A be a complex square matrix, then $A + A^*$ is Hermitian and $A - A^*$ is skew-Hermitian.

<u>Pf</u>:- HW

<u>Th</u>: A complex square matrix can be uniquely expressed as the sum of a Hermitian matrix and a skew-Hermitian matrix.

<u>Pf</u>: Let A be a given matrix, then A can be expressed as

$$A = \frac{1}{2}(A + A^*) + \frac{1}{2}(A - A^*).$$

$$\left[\frac{1}{2}(A + A^*)\right]^* = \frac{1}{2}\left(A^* + (A^*)^*\right) = \frac{1}{2}(A + A^*).$$

$$\left[\frac{1}{2}(A - A^*)\right]^* = \frac{1}{2}(A^* - A) = -\frac{1}{2}(A - A^*)$$

$\square$

Unitary Matrix

Def: A complex matrix is said to be unitary if $AA^* = I$

Ex- $\begin{pmatrix} \frac{1+i}{2} & \frac{1-i}{2} \\ \frac{1-i}{2} & \frac{1+i}{2} \end{pmatrix}$, $\frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}$ where $\omega = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$.

NB: For a real unitary matrix A, $AA^* = AA^t = I$.
Real unitary matrix is a real orthogonal matrix.

Prop (1) A be a unitary matrix then

(i) $|\det(A)| = 1$

(ii) $A^*A = I$

(iii) $A^{-1} = A^*$

Pf: ~~(i) $|AA^{-t}| = |I| = 1$~~

~~$\Rightarrow |A$~~

(i) $AA^* = I$

$\Rightarrow \det(AA^t) = \det(I) = 1$

$\Rightarrow \det(A)\det(\bar{A}) = 1$

$\Rightarrow \det(A).\overline{\det(A)} = 1$

$\Rightarrow |\det(A)|^2 = 1$

$\Rightarrow |\det(A)| = 1$

(ii) & (iii) Since A^{-1} exists ~~and~~ A is a square matrix, and $AA^* = I$

$\Rightarrow A^{-1} = A^*$ and $A^*A = I$

(NB: A is non singular, So A^{-1} exists.)

(ii) $AA^* = I \Rightarrow (AA^*)A = IA$

$\Rightarrow A(A^*A) = AI$

$\Rightarrow A(A^*A - I) = 0$

$\Rightarrow A^{-1}\{A(A^*A - I)\} = 0$ $\quad \because A$ is unitary $\therefore A^{-1}$ exists

$\Rightarrow A^*A = I$

(iii) $A^*A = I = AA^*$ (by def.)

$\Rightarrow A^{-1} = A^*$

Prop ② ~~(i)~~ $A_{n\times n}, B_{n\times n}$ are unitary then

(i) AB is also unitary

(ii) A^{-1} is also unitary.

Pf:- HW

Eigenvalues and eigenvectors

The characteristic equation of a matrix

Let A be a $n \times n$ matrix (over a field F).

Characteristic polynomia : ~~det(A − x)~~ $\det(A - xI_n) =: \psi_A(x)$

Charateristic equation : $\psi_A(x) = 0$

Ex :- ① $A = \begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix}$.

$$\psi_A(x) = \begin{vmatrix} 2-x & 1 \\ 3 & 5-x \end{vmatrix} = x^2 - 7x + 7 \quad \text{characteristic polⁿ.}$$

Then the characteristic eqⁿ : $x^2 - 7x + 7 = 0$

② Let $A = (a_{ij})$ then $\psi_A(x) = \begin{vmatrix} a_{11}-x & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22}-x & \cdots & a_{2n} \\ \vdots & & & \\ a_{n1} & a_{n2} & \cdots & a_{nn}-x \end{vmatrix}$

$$= c_0 x^n + c_1 x^{n-1} + \cdots + c_n$$

by deleting i_1th, i_2th, ..., i_{n-r}th rows and the i_1th, i_2th, ..., i_{n-r}th columns

Here $c_0 = (-1)^n$

$c_r = (-1)^{n-r}$. (sum of all the principal minors of A of order r)

$\therefore c_1 = (-1)^{n-1}(a_{11} + a_{22} + \cdots + a_{nn}) = (-1)^{n-1}\,\text{trace}(A)$

$c_n = \det(A)$.

Cayley - Hamilton theorem:

Every square matrix satisfies its own characteristic equation.

- If $\psi_A(x) = 0$ is the characteristics eqn of A then $\psi_A(A) = 0$

i.e., $c_0 A^n + c_1 A^{n-1} + \cdots + c_n I = 0$

Ex- $A = \begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix}$

$\psi_A(x) := x^2 - 7x + 7 = 0$

So by Cayley - Hamilton theorem: $A^2 - 7A + 7I = 0$

$$= \begin{pmatrix} 7 & 7 \\ 21 & 28 \end{pmatrix} - \begin{pmatrix} 14 & 7 \\ 21 & 35 \end{pmatrix} + \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = 0$$

Note: If $|A| \neq 0 \Rightarrow c_n \neq 0$

Now $\psi_A(A) = 0 \Rightarrow c_0 A^n + c_1 A^{n-1} + \cdots + c_n I_n = \theta$

$$\Rightarrow -c_n^{-1}(c_0 A^{n-1} + c_1 A^{n-2} + \cdots + c_{n-1} I_n) A = I_n$$

$$\therefore A^{-1} = -c_n^{-1}(c_0 A^n + \cdots + c_{n-1} I_n)$$

Ex - $A = \begin{pmatrix} 2 & 1 \\ 3 & 5 \end{pmatrix}$

By $\psi_A(x) = x^2 - 7x + 7 = 0$

$\Rightarrow A^2 - 7A + 7I = 0$ Here $c_n = -7 \neq 0$

$\Rightarrow -\frac{1}{7} A (A - 7I_2) = I_2 \Rightarrow A^{-1} = -\frac{1}{7}(A - 7I_2)$

$$= -\frac{1}{7}\begin{pmatrix} -5 & 1 \\ 3 & -2 \end{pmatrix}$$